TECHNICAL ARCHITECTURE & SYSTEM DESIGN

Campus Marketplace

AI Pricing & Fraud Detection for a Verified Student Trading Ecosystem

MERN Stack | Python / FastAPI ML Services | MongoDB Atlas | Geospatial Discovery

Academic Project Proposal
August 2026

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1. What We Are Building & Why

Campus Marketplace is a campus-exclusive platform where verified students buy, sell, lend, and borrow used items such as textbooks, cycles, electronics, lab gear, furniture, notes, and hostel items.

It differs from a generic classifieds application in three ways:

1. Access is restricted to verified institutional email holders.
2. An AI pricing engine tells buyers and sellers whether an asking price is fair.
3. A fraud-risk model screens listings and accounts before problems reach students.

The extended system additionally provides:

- **Borrow/Lend Due-Date Tracking** for temporary item transactions.
- **Price-Drop Alerts** for buyers watching an item.
- **Trust Score** based on verified activity, transaction history, reviews, and reports.

The goal is a safe, transparent, AI-assisted trading ecosystem—not a full e-commerce clone.

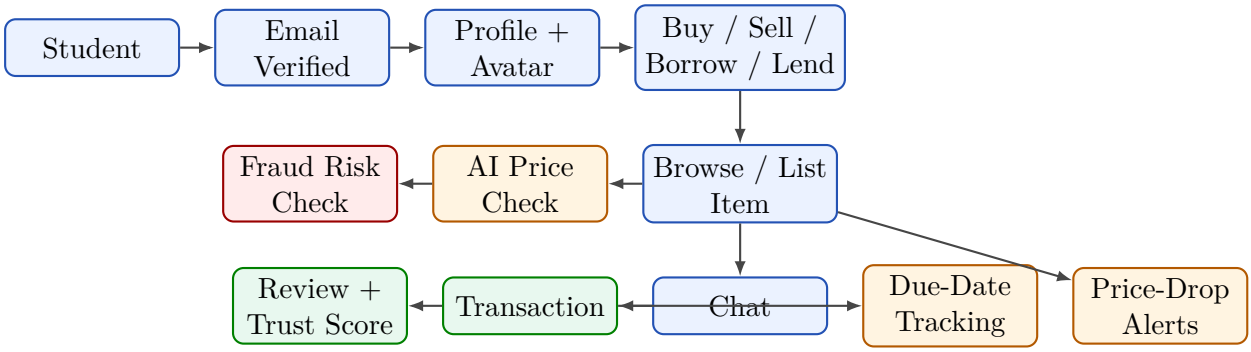


Figure 1: High-level end-to-end system flow. AI pricing and fraud checks run in parallel with normal marketplace activity.

2. User Roles

Students are not locked into a fixed role. After login, they choose an action from the dashboard.

Role	Capabilities
Student	Buy, sell, borrow, lend, browse, chat, transact, review users, watch listings, receive price-drop alerts.
Borrower	Request borrowed items, accept terms, view due dates, receive return reminders, mark items returned.
Lender	List items for lending, set lending duration/due date, approve borrowers, track active and overdue loans.
Admin	Institution configuration, monitor listings, review fraud alerts, handle reports, suspend users, remove listings, analytics, and banned-word/category management.

3. College Email Verification



Domain matching confirms institutional affiliation only, not trustworthiness. Email verification is therefore combined with CAPTCHA/rate limiting, duplicate-account detection, report/block tools, and admin moderation.

3.1 Avatar-Based Identity

Users build an identity from predefined components such as face, hair, clothing, accessories, and colour. These are stored as a small JSON object in the `Users` collection. Real photos are not required, reducing onboarding friction and the safety/moderation burden.

AI-generated avatars are left as a future enhancement.

4. Optional Location-Based Discovery

Location sharing is off by default and fully opt-in. Exact coordinates are never shown publicly. Only approximate/geohashed positions are used for discovery, while precise meeting points are revealed only after both parties agree.



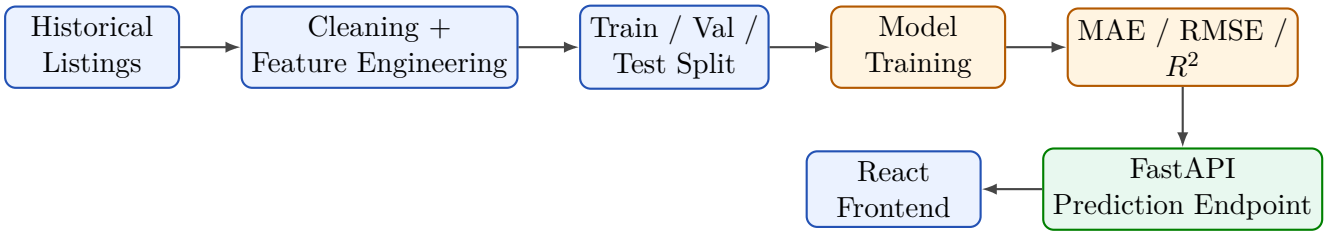
MongoDB's `$geoNear`/`$near` queries support nearby searches directly on a geospatial index.

Safeguards:

- Disable location sharing at any time.
- Never publish exact coordinates.
- Use geohash rounding for approximate discovery.
- Reveal precise location only after mutual agreement.
- Encrypt stored location data.
- Automatically expire stale location data.

5. AI Price Recommendation

Rather than returning a single number, the model returns a fair price range, confidence score, and plain-language reason. This helps both sellers and buyers assess whether an asking price is reasonable.



Features: original price, age, condition, brand, category, demand, and prior campus listings.

Model	Recommendation
Linear Regression	Good simple baseline; may struggle with nonlinear condition and brand effects.
Random Forest Regressor	Recommended for v1; handles mixed numeric/categorical features well and is robust on a small campus dataset.
Gradient Boosting / XGBoost	Potential v2 upgrade once more labelled data accumulates.

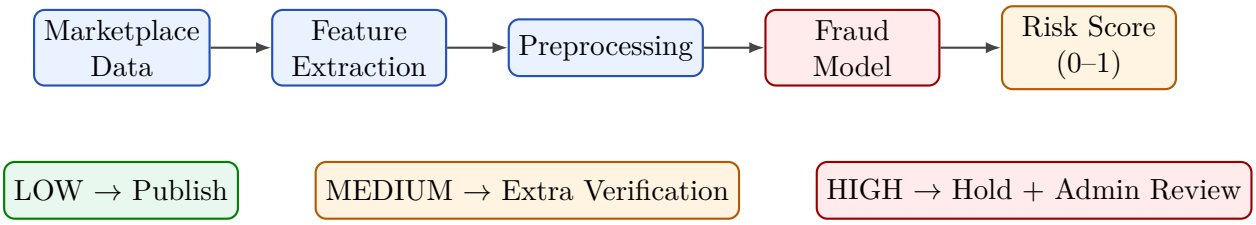
Accuracy is intentionally not overclaimed: outputs are framed as an "AI-estimated fair range", not a guaranteed price.

6. Fraud / Scam Detection

Detection operates at three levels:

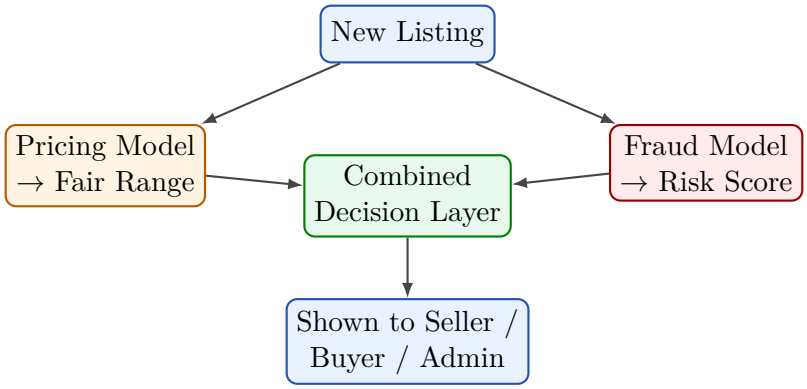
- **Listing:** price outliers, reused images, duplicate posts.
- **User:** multiple accounts, rapid listing bursts, prior reports.
- **Behavioural:** unusual activity spikes.

The system produces a single **Fraud Risk Score** that flags suspicious activity rather than automatically accusing a user.



The recommended initial approach is Isolation Forest (unsupervised anomaly detection), since labelled fraud data will not exist initially. Logistic Regression or Random Forest can be introduced later when sufficient confirmed-fraud labels are collected from admin reviews.

7. Combined AI Decision Engine



For example, an iPhone 13 listed 30% below the estimated market range does not automatically become a fraud case. Price deviation is only one signal among several weighted by the risk model.

8. Borrow/Lend Due-Date Tracking

Borrowing and lending are treated as first-class transaction types rather than simple sales.

8.1 Borrow/Lend Workflow

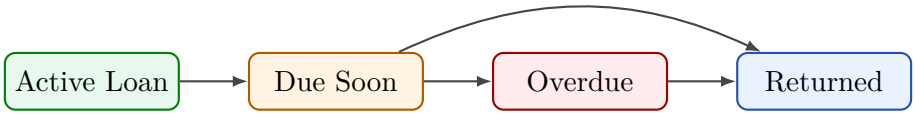


When a lender approves a borrow request, the transaction stores:

- Borrower and lender IDs.
- Start date.
- Due date.
- Optional extension count and extension history.
- Return status.
- Overdue status.

The dashboard displays active loans and their remaining time. Automated reminders can be generated before the due date and when an item becomes overdue.

8.2 Due-Date States



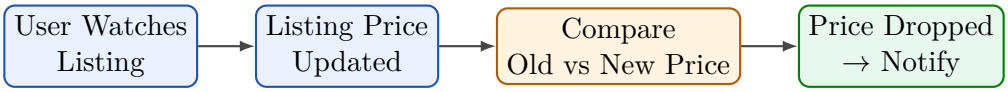
Business rules:

- A due date is mandatory for borrow/lend transactions.
- Borrowers receive a reminder before the due date.
- Overdue transactions are visible to both parties and the admin system.
- Repeated overdue returns can reduce a user's Trust Score.
- A lender may accept or reject an extension request.
- A transaction can be marked complete only after the lender confirms return.

9. Price-Drop Alerts

Users can save/watch a listing and choose to receive an alert when its price decreases.

9.1 Alert Logic



The alert service stores the user's watch preferences and the last observed price. When the seller lowers the listed price, the system generates an in-app notification and can optionally send an email notification.

A basic rule is:

$$\text{Price Drop} = \begin{cases} 1, & P_{\text{new}} < P_{\text{old}} \\ 0, & \text{otherwise} \end{cases}$$

Optional future refinement can trigger alerts only when the reduction exceeds a user-selected threshold such as 5% or a fixed amount.

10. Trust Score

The **Trust Score** provides a compact representation of a user's marketplace reliability. It is not the same as the AI Fraud Risk Score: fraud risk evaluates suspicious behaviour, whereas Trust Score represents accumulated positive and negative transaction history.

10.1 Trust Score Inputs

The score can consider:

- Verified institutional account.
- Successful completed transactions.
- Average review rating.
- Number of positive reviews.
- On-time return rate for borrowed items.
- Number and severity of confirmed reports.
- Overdue borrow/lend history.
- Account age and normal marketplace activity.

A simple normalized formulation for the initial implementation is:

$$T = w_v V + w_r R + w_c C + w_o O - w_f F$$

where:

- V = verification component,
- R = review/rating component,
- C = completed transaction component,
- O = on-time return component,
- F = confirmed negative-report component,
- w_i = configurable weights.

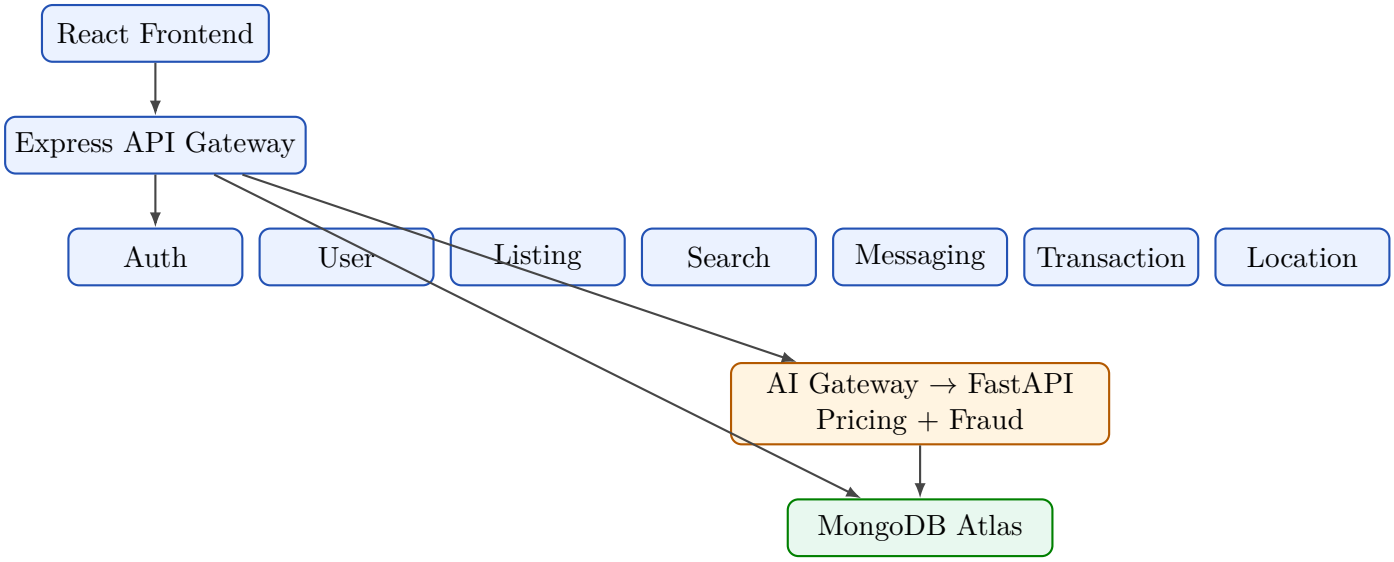
The final score is normalized to a 0–100 range.

Score Range	Interpretation
80–100	Highly trusted marketplace history
60–79	Good / established history
40–59	Limited or mixed history
0–39	Low trust; additional caution or verification may be appropriate

The score should be presented as a decision-support indicator rather than a guarantee of user behaviour. Confirmed reports should affect the score more strongly than unverified complaints.

11. Backend Architecture

A modular monolithic Node.js/Express backend with logically separated modules, plus independent Python/FastAPI ML services, is recommended over full microservices. This is simpler to build, test, and deploy for a project of this scope while keeping AI logic decoupled.



12. Database Schema (MongoDB)

Collection	Key Fields
Users	userId, collegeEmail, avatar, role/preferences, verificationStatus, trustScore, locationSharingEnabled, approximateLocation, createdAt
Listings	listingId, sellerId, title, category, condition, images, listedPrice, AIRecommendedPrice, fairPriceRange, fraudRiskScore, listingType (sale/borrow/lend), location, alertEligible
Transactions	transactionId, buyerId/borrowerId, sellerId/lenderId, listingId, type, status, startDate, dueDate, returnedAt, extensionCount, overdueStatus, createdAt, completedAt
Messages	messageId, senderId, receiverId, listingId, message, timestamp
Reviews	reviewerId, reviewedUserId, transactionId, rating, comment
FraudReports	reportId, reporterId, reportedUserId/listingId, reason, riskScore, status
Watchlists	watchId, userId, listingId, lastObservedPrice, alertEnabled, thresholdType, thresholdValue, createdAt
Notifications	notificationId, userId, type, listingId/transactionId, message, readStatus, createdAt

`Listings.location` uses a GeoJSON Point with a 2dsphere index to support nearby-item queries directly in MongoDB.

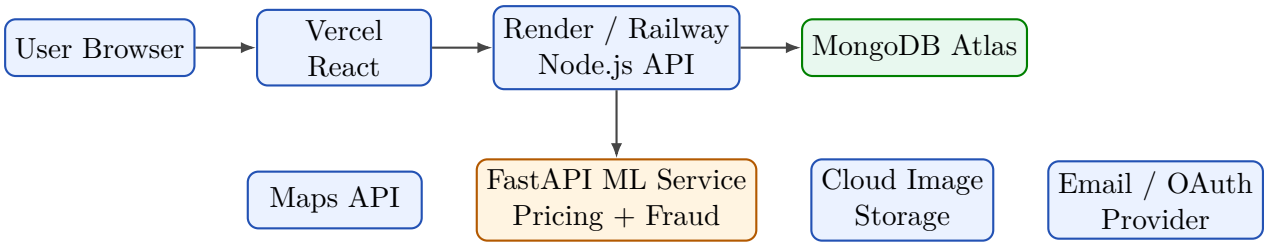
13. Key APIs

Domain	Endpoints
Auth	POST `/auth/register`, `/verify`, `/login`, `/logout`
Listings	POST/GET/PUT/DELETE `/listings[/:id]`
AI	POST `/ai/price-predict`, POST `/ai/fraud-score`
Location	PUT `/location/preferences`, PUT `/location`, GET `/location/nearby`
Messages	POST `/messages`, GET `/messages/:conversationId`
Transactions	POST `/transactions`, PUT `/transactions/:id/due-date`, PUT `/transactions/:id/return`, POST `/transactions/:id/extend`
Watchlist	POST `/watchlist`, DELETE `/watchlist/:listingId`, GET `/watchlist`
Alerts	GET `/notifications`, PUT `/notifications/:id/read`
Trust	GET `/users/:id/trust-score`
Admin	GET `/admin/fraud-alerts`, PUT `/admin/listings/:id/review`

14. Security

- College email verification.
- Hashed passwords and JWT sessions.
- HTTPS.
- Input validation and rate limiting.
- CORS configuration.
- Role-based access control.
- Signed image uploads with size/type checks.
- Duplicate-account detection.
- Geohashed location storage; exact location is never public by default.
- Chat abuse reporting.
- Admin-only authenticated routes.
- Audit trail for trust-score-impacting actions.

15. Deployment Architecture



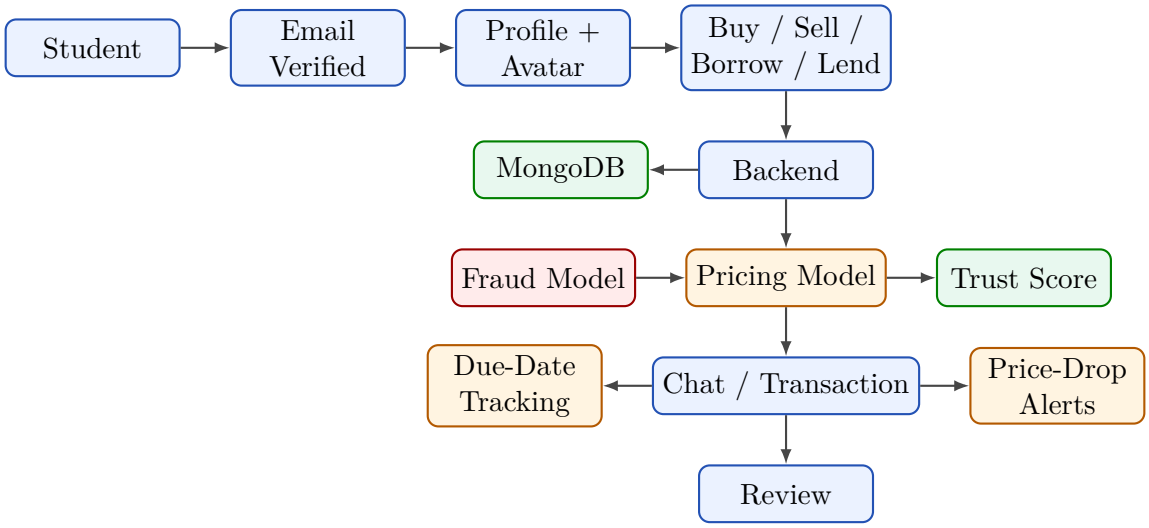
A practical cost-realistic deployment uses managed platforms for the frontend, backend, database, and independently deployed ML service.

Mapbox is recommended for a flexible campus map and a student-friendly deployment model.

16. MVP vs. Future Enhancements

Feature	Status
Email authentication, profiles, avatars, listings, search, messaging, reviews	MVP
AI price recommendation using Random Forest	MVP
Basic fraud scoring using Isolation Forest	MVP
Optional approximate location + admin moderation	MVP
Borrow/Lend due-date tracking and overdue reminders	MVP
Price-drop alerts and listing watchlist	MVP
User Trust Score	MVP
Mobile app	Future
Computer-vision fraud detection	Future
OCR invoices	Future
AI chat assistant	Future
Escrow/payments	Future
Demand forecasting	Future

17. Final System Pipeline



The complete pipeline combines verified student access, marketplace functionality, AI-assisted price estimation, fraud risk scoring, optional geospatial discovery, due-date tracking, price-drop notifications, and trust mechanisms.

18. Summary

Campus Marketplace combines a verified campus community, marketplace functionality, AI-assisted price estimation, fraud risk scoring, optional location-based discovery, and trust mechanisms into one ecosystem.

The added features make the platform more useful beyond simple buying and selling:

- **Borrow/Lend Due-Date Tracking** makes temporary transactions manageable and reduces missed returns.
- **Price-Drop Alerts** help buyers discover when a watched item becomes cheaper.

- **Trust Score** gives students a quick view of accumulated marketplace reliability.

The AI layer estimates and flags; it does not guarantee fair pricing or eliminate fraud outright. Final decisions on flagged listings remain with admin review.

Campus Marketplace — Technical Architecture Document
Prepared as an academic project proposal — August 2026